

Transformers, Attention and Pre-training

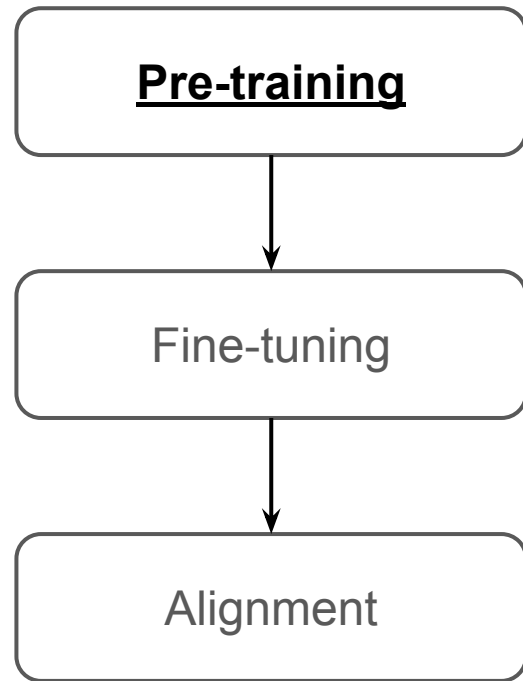
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Notebook: <https://tinyurl.com/4fnu388x>

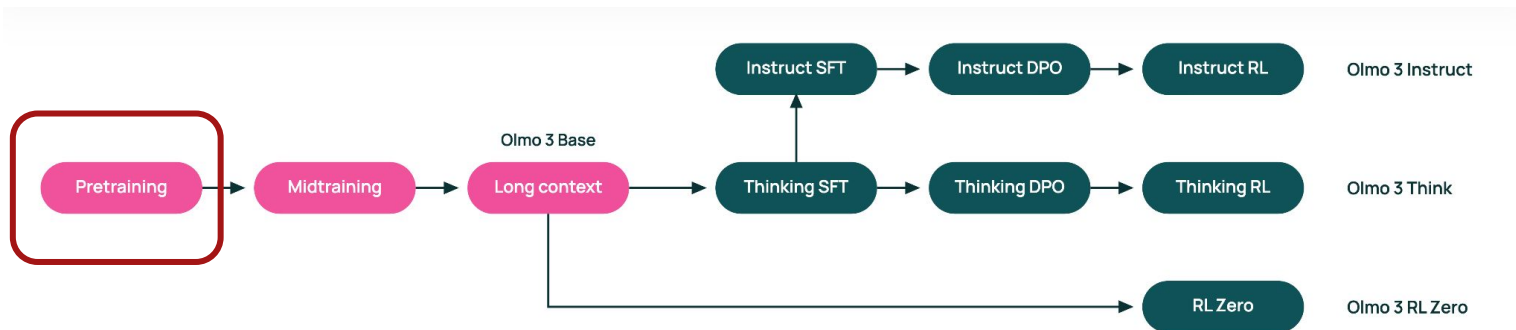


What is pre-training?

- LLMs are capable of many tasks that can be represented and processed in natural language.
- But before they can do these tasks, they must first be *fluent* in natural language.
- **Pre-training** is the process of making an LLM fluent in human languages.
- This process is called **language modeling**. (What??)



What is pre-training?



What does it mean to *model* language?

- In the beginning the universe was created.
- Happy families are all alike; every unhappy family is unhappy in its own way.
- Intelligent life on a planet comes of age when it first works out the reason for its own existence.
- Colorless green ideas sleep furiously.
- Contextual nuances highlight the recursive tension inherent to the framework.
- True value lies in embracing the multifaceted tapestry of perspectives.

To **model** language is to have a way of telling how **plausible** a sentence would arise from an actual human language.

We all have an (informal) model of language in our heads.

Probability

- Given a sentence s = “In the beginning the universe was created.”
- What is its probability, $P(s)$?
- Ideally:
 - A human-generated sentence would have high $P(s)$.
 - Other sentences would have much lower $P(s)$.
- So, how we we find this probability?

Sequence of tokens

- A sentence (or a paragraph, a passage, etc.) is represented as a **sequence**.
- Each element in a sequence is a **token**.
- The process of breaking down a sentence is called **tokenization**.

So s = “In the beginning the universe was created.” would be tokenized into

In the beginning the universe was created .

which is a sequence of 8 tokens.

Causal language modeling

Generative process: one token at a time, from left to right.

$$\begin{aligned} & P(\text{In the beginning the universe was created} \cdot) \\ &= P(\text{In}) \\ &\times P(\text{the} \mid \text{In}) \\ &\times P(\text{beginning} \mid \text{In the}) \\ &\dots \\ &\times P(\cdot \mid \text{In the beginning the universe was created}) \end{aligned}$$

Causal language modeling

- So, for a sequence $s = (X_1, X_2, \dots, X_n)$
- The joint probability of this sequence is

$$P(s) = \prod_{i=1}^n P(X_i \mid X_{1:i-1})$$

- If $i = 1$, then $X_{1:i-1} = \emptyset$, so $P(X_i \mid X_{1:i-1}) = P(X_1)$
- So, one marginal probability, and $n - 1$ conditional probabilities.

Causal language modeling

Our language model P can

- Estimate the probability of any sequence. The closer it is to human language, the higher its probability.
- **Also generate:** Given any context s , $P(\cdot | s)$ estimates the probability of *any* possible next token. We can thus sample from $P(\cdot | s)$ “indefinitely.”

Most modern LLMs are causal sequence models.

Generating text from a causal language model

Original abstract

The Magellanic Stream (MS) - an enormous ribbon of gas spanning 140° of the southern sky trailing the Magellanic Clouds - has been exquisitely mapped in the five decades since its discovery. However, despite concerted efforts, no stellar counterpart to the MS has been conclusively identified. This stellar stream would reveal the distance and 6D kinematics of the MS, constraining its formation and the past orbital history of the Clouds. We have been conducting a spectroscopic survey of the most distant and luminous red giant stars in the Galactic outskirts. From this dataset, we have discovered a prominent population of 13 stars matching the extreme angular momentum of the Clouds, spanning up to 100° along the MS at distances of 60–120 kpc. Furthermore, these kinematically-selected stars lie along a $[\alpha/\text{Fe}]$ -deficient track in chemical space from $-2.5 < [\text{Fe}/\text{H}] < -0.5$, consistent with their formation in the Clouds themselves. We identify these stars as high-confidence members of the Magellanic Stellar Stream. Half of these stars are metal-rich and closely follow the gaseous MS, whereas the other half are more scattered and metal-poor. We argue that the metal-rich stream is the recently-formed tidal counterpart to the MS, and speculate that the metal-poor population was thrown out of the SMC outskirts during an earlier interaction between the Clouds. The Magellanic Stellar Stream provides a strong set of constraints - distances, 6D kinematics, and birth locations - that will guide future simulations towards unveiling the detailed history of the Clouds.

Completed by GPT-4

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Completed by LLaMA-2

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Completed by AstroLLaMA

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Causal language modeling

How do we find $P(\text{In})$?

- Go out there and find all sentences in English.
- Count the number of times a sentence starts with In .
- Normalize this count to get an estimate of $P(\text{In})$.

Causal language modeling

How do we find $P(\text{the} \mid \text{In})$

- Go out there and find all sentences in English.
- Count the number of times a sentence starts with `In`.
- Count the number of times such a sentence is followed by `the`.
- Normalize to get the condition probability.

Problem!!

- Let's say the English language has about 100,000 words.
- For a sequence of **2** words
 - We need to store every combination of X_1 and X_2 .
 - So we need to store **100,000²** probabilities
- For a sequence of **n** words
 - We need to store **100,000ⁿ** probabilities!!

Solution: Train a statistical model such much fewer parameters.

Evaluating a language model

- Any causal language model has the form $P(\cdot | \cdot)$.
- How do we evaluate competing models of our data? Likelihood.
- For a sequence s , $P(s)$ is the likelihood of our model.
- Thus “better” models would give higher $P(s)$ for each of our sentences.

Evaluating a language model

- Recall that

$$P(s) = \prod_{i=1}^n P(X_i \mid X_{1:i-1}) = P(X_1) \times P(X_2 \mid X_1) \times \dots \times P(X_n \mid X_{1:n-1}).$$

- **Problem!!!** Longer sequences tend to have much lower likelihoods.

Accounting for sequence length

- We need to normalize likelihood by sequence length.
- First, turn likelihood into log-likelihood

$$\log P(s) = \sum_{i=1}^n \log P(X_i \mid X_{1:i-1})$$

- Second, divide by the number of tokens

$$\frac{1}{n} \log P(s) = \frac{1}{n} \sum_{i=1}^n \log P(X_i \mid X_{1:i-1})$$

- Third, exponentiate its negative, and we have the **perplexity score**

$$\text{PPL}(s) = \exp \left(-\frac{1}{n} \log P(s) \right).$$

Training a language model

$$P(\text{next_token} \mid \text{some context})$$

- Estimating the conditional $P(\cdot \mid \cdot)$ is doable; we just need to count.
- But it is *not* feasible due to the complexity of the space.
- Instead, we will train a model to approximate $P(\cdot \mid \cdot)$.
- What kinds of model can do this well?

N-gram language model

- Markov assumption: ignore the entire context but the latest (N-1) tokens.

$$P(\text{beginning} \mid \text{In the}) = P(\text{beginning} \mid \text{the})$$

- With 2-grams, complexity is now just **100,000²** probabilities.
- Easy to understand and calculate, fully interpretable.
- Longer context tends to fit better. But overall does not generalize very well.

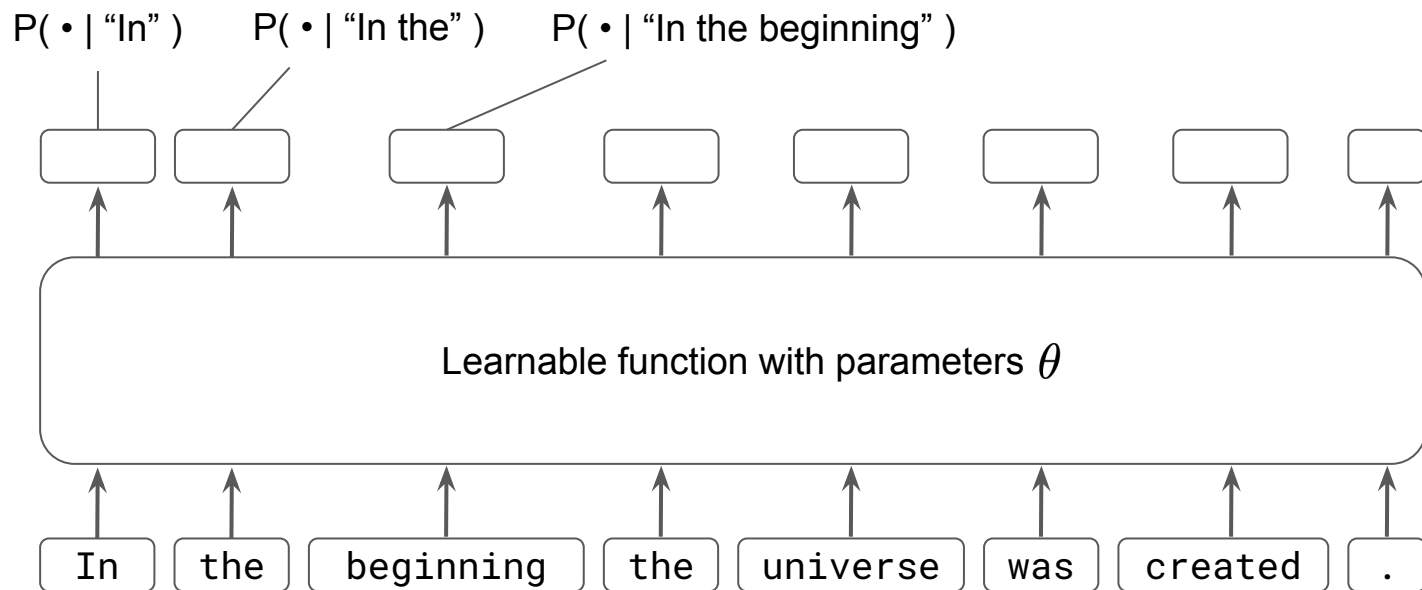
	Unigram	Bigram	Trigram
Perplexity	962	170	109

Jurafsky & Martin 2026, *Speech and Language Processing*, 3rd ed., p. 10.

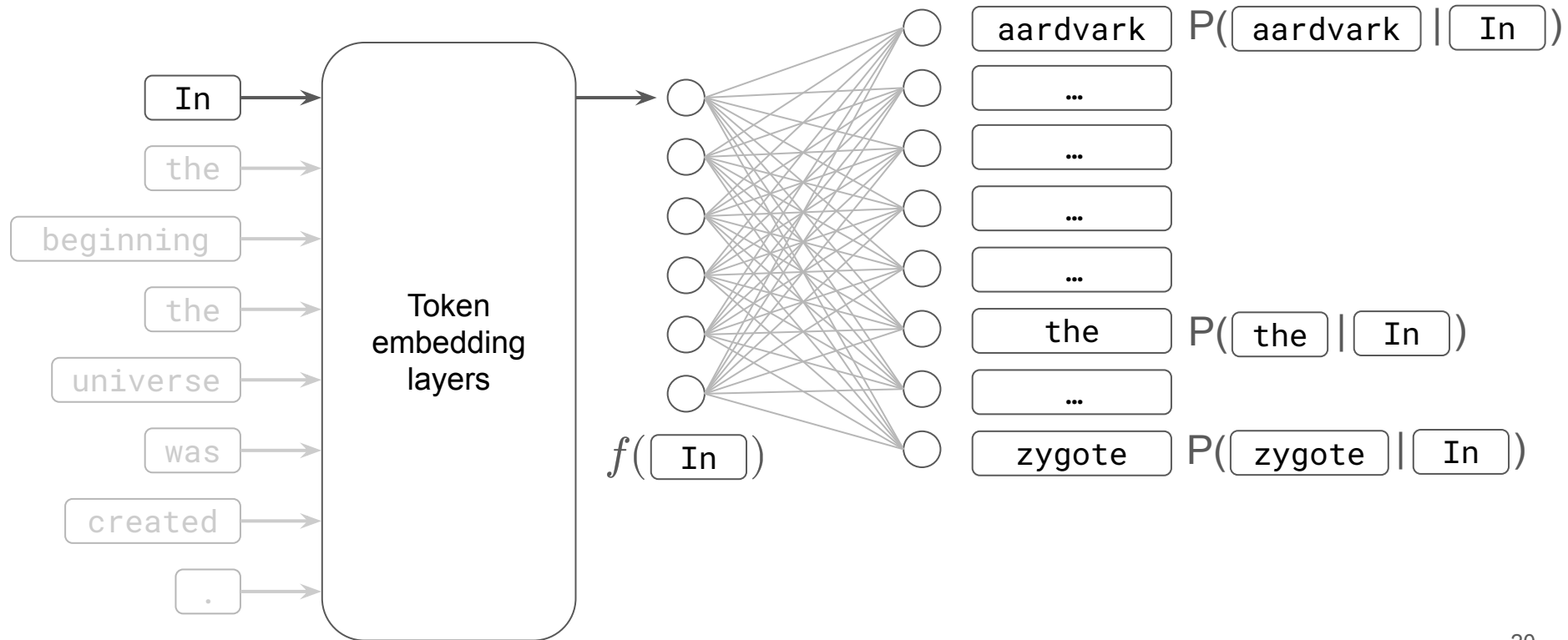
N-gram models trained on 38M words from WSJ

Neural network language model

- Train a model P that approximates prob. of the next token, $P(\cdot | \cdot)$.
- Find model parameters θ that minimize the perplexity of the training set.



Predicting the next token



Embedding a token

- The embedding function f
 - Takes any token in our vocabulary, like In
 - Then transforms it into a dense vector $f(\text{In})$
- The crux of our language model is to design a good embedding function.
- Several decisions
 - Number of dimensions: 512, 768, 1024, 2048, etc.
 - How it is composed and learned: feedforward, convolution, recurrent, attention, etc.

Attention: choosing the right context

Imagine generating this sentence

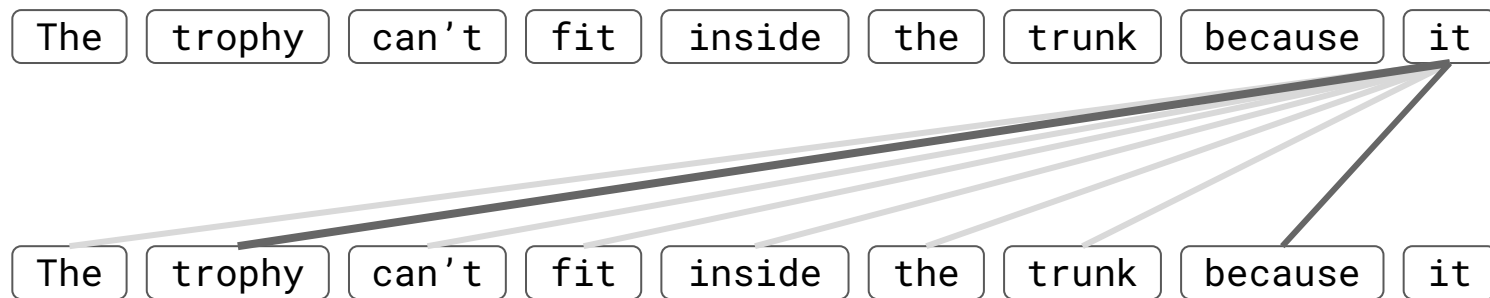
The trophy can't fit inside the trunk because it

How should we determine the embedding of `it` ?

- `it` is related to certain words than others.
- Namely, `trophy` and `trunk`, depending on the content.
- Either way, `it` should mostly “attend” to `trophy` and `trunk`.

Attention: choosing the right context

If we were to display it like a graph, it would be something like this



So, if the embedding of `it` may rely on the context before it, the parts of that context that matter the most are `trophy` and `trunk`.

Attention: choosing the right context

Computationally, the attention “head” has three parametrized components

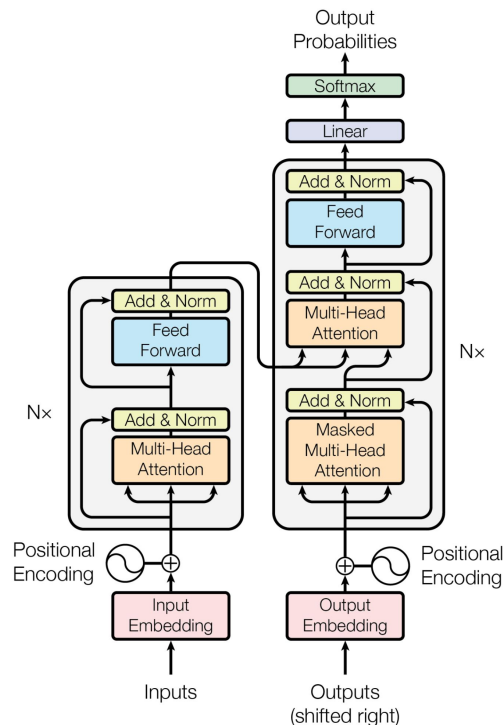
- **Query**: a vector that represents `it`.
- **Key**: a set of vectors that represent the context, including `trophy` and `trunk`.
- **Value**: a set of vectors for all words to compute the output.

The final embedding of `it` is the weighted average of all **values**, the weights being determined by the similarity between the **query** and **keys**.

$$A = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Transformer block

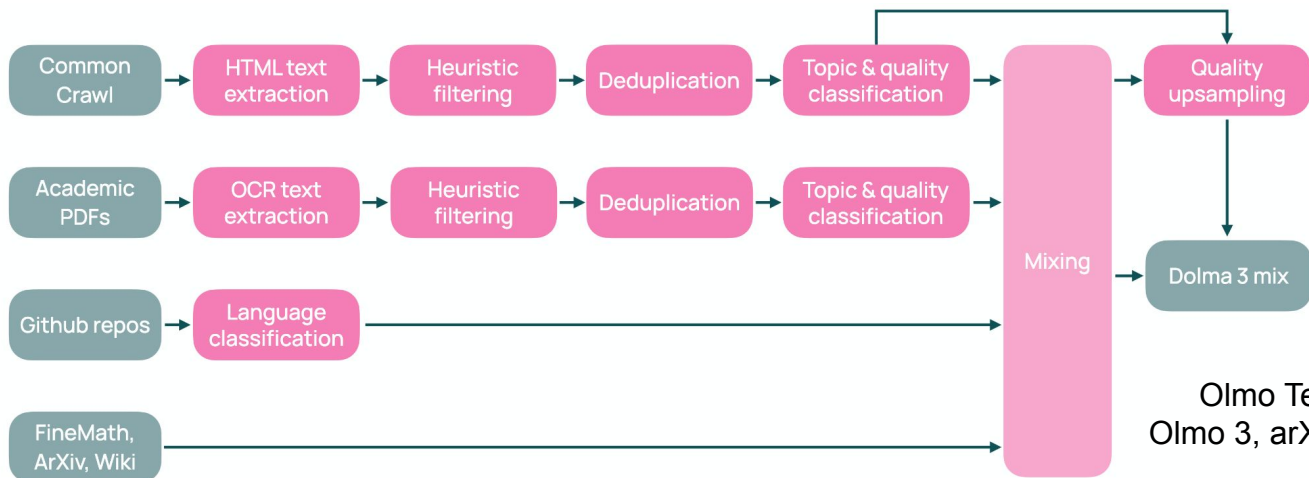
- A transformer block includes **attention** and other other components, such as **residual** stream, layer **norm** and **feedforward**.
- Each token “attends” to everything else before it, just with different weights.
- Extremely useful for sequence modeling.



Vaswani *et al.* 2017, Attention Is All You Need, arXiv: 1706.03762

Pre-training in practice

- Find model parameters that minimize perplexity on a training set.
- Huge engineering and computational endeavor. Done mostly by big players.
- Lots of data quality control needed.



Olmo Team *et al.* 2026,
Olmo 3, arXiv: 2512.13961

Continual pre-training

- Pre-training from scratch is most likely beyond our means.
- But **continual pre-training** can be.
- Idea: take a pre-trained LLM, continue training it on our dataset.
- Goal: make the model more **fluent** in the our domain than it already is.

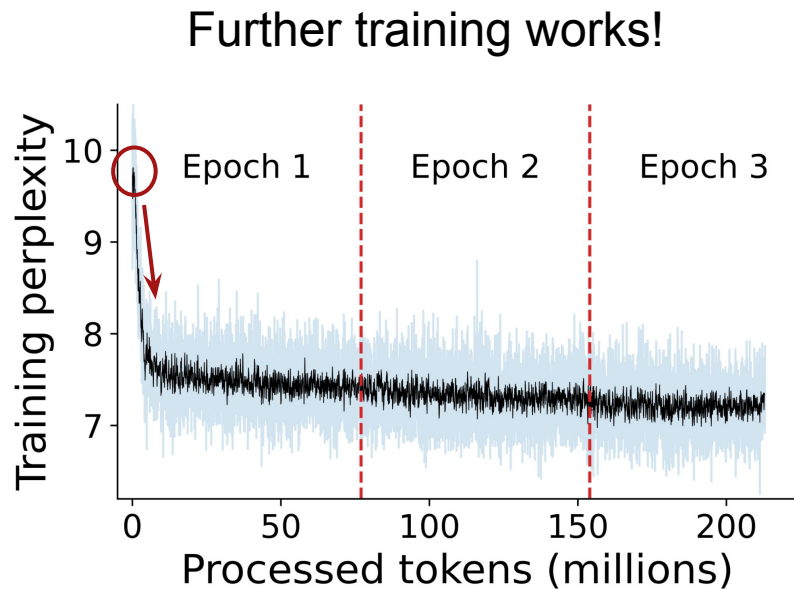
AstroLLaMA : Towards Specialized Foundation Models in Astronomy

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Charles O'Neill^{2†}, Ze-Chang Sun^{4†}, Maja Jabłońska^{2†}, Sandor Kruk^{5†}
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arXiv: 2309.06126

Continual pre-training

- 326,238 abstracts from arXiv astro-ph (1992-04 to 2023-07).
- 95M tokens, 77M used for training.
- Pre-trained model: LLaMA-2, 6.7B parameters.
- Model fit evaluated by perplexity.



“Extrinsic” evaluation

- Perplexity measures model fit to our data, regardless of the task. Thus it is called an **intrinsic metric**.
- But models must also be evaluated **extrinsically**, i.e., based on tasks that we actually use them for.
- What kinds of task can we do with a pre-trained LLM?
 - Generating text
 - Embedding text for similarity and retrieval

Generating more plausible text

Original abstract

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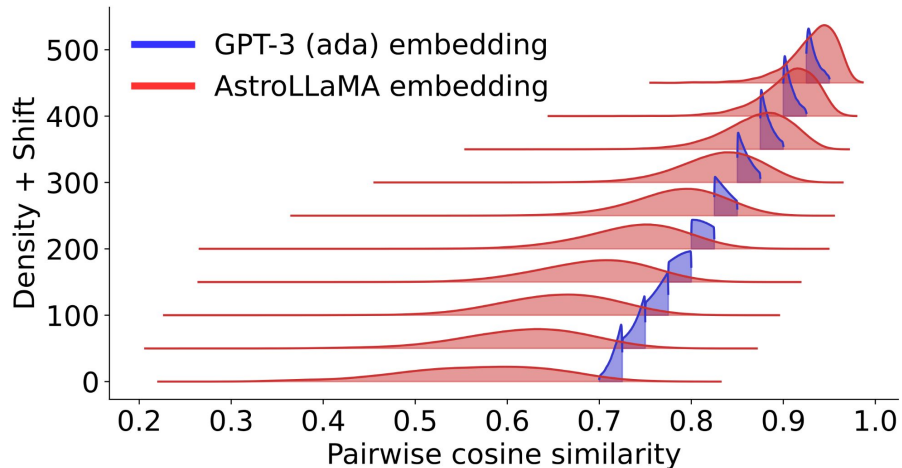
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Embedding texts for retrieval

- We used 2 LLMs to embed 10,000 abstracts using their internal vector representation of each token.
- For each pair of abstracts, we calculated their cosine similarity.
- **GPT** thinks most abstracts are **very similar** (domain generality).
- **AstroLLaMA** thinks there's **much more variance** in similarity (suggesting domain specialization).



Paper 1: Astrophysical gyrokinetics: kinetic and fluid turbulent cascades in magnetized weakly collisional plasma

Paper 2: Comment on modified Coulomb law in a strongly magnetised vacuum

GPT-3 cosine similarity score: **78.5%**

AstroLLaMa cosine similarity score: 36.3%

Paper 1: A Spitzer census of the IC 348 nebula

Paper 2: Sequential and spontaneous star formation around the mid-infrared halo HII region KR 14

GPT-3 cosine similarity score: 82.4%

AstroLLaMa cosine similarity score: **92.8%**